



Meena Radiology Scheduling

Project Overview

An all-in-one platform for the radiology department's monthly rota, approvals, leave, swaps and daily case reports.

1. Overview

Meena Radiology Scheduling is a single web platform that runs the whole scheduling life-cycle for the radiology department across all branches (NESTS): building the monthly rota, approvals, leave, shift swaps, and the daily case reports — replacing scattered Excel files and chat groups.

Goal: turn scheduling into one organised, secure, fast system with clear permissions for every role.

2. The problem before

- Excel rotas edited by hand — repeated mistakes.
- Leave and swap requests lost in chat threads.
- No record of who approved what, and when.
- Hard to see each branch's daily cases.
- Reports built manually, costing time.

3. The solution — key features

Feature	What it does
Auto-generate rota	A smart engine (OR-Tools) builds the monthly rota respecting shifts, approved leave and limits.
Clear approval chains	Schedule: create -> review -> approve -> lock. Leave: staff -> team lead -> manager.
Leave on the rota	Approved leave lands on the schedule automatically, with a warning if it leaves a shift uncovered.
Shift swaps	Colleague -> team lead -> manager approval, then applied to the rota automatically.
Daily case reports	Each branch logs its cases (X-ray/CT/US/...); exported as a branded PDF
Self sign-up	Role-based invite links (staff / team lead / manager) with approval.
Notifications + email	In-app and email alerts on every important step.
Professional PDFs	Rota and case reports in Meena's brand, ready to print and archive.

4. Roles & permissions

Role	Can do
Super Admin	Full control: branches, shift types, users, audit log, invite links, data reset.
Manager	Reviews & approves schedules, final leave approval, sees all branches.
Team Lead	Runs their own branch: build/generate rota, first leave approval, approve their staff sign-ups.
Staff	Personal portal: own schedule, request leave, request swaps, fill cases (if eligible).
Viewer	Read-only.

Permissions are branch-scoped: a team lead can't see or edit another branch's data.

5. Security

- Passwords hashed (bcrypt); sessions secured with JWT.
- Idle auto-logout and safe session expiry.
- Audit log of every sensitive action: who did what, when.
- Every permission enforced on the server (not just the UI).
- 240+ automated tests guarding the permissions and flows.

6. Benefits for management

- Time saved: the rota is built in one click instead of hours.
- Governance: every approval is recorded with a name and date.
- Live visibility: each branch's cases, the rota and leave in one dashboard.
- Ready reports: branded PDF for printing and official archiving.
- Works on phone and desktop.

7. Suggested next steps

- Optional WhatsApp notifications.
- Monthly aggregated performance reports.
- Historical case-statistics dashboard.